

CY 201P : Physical Chemistry Laboratory

Course Code : CY 201P

Course Name : Physical Chemistry Laboratory

L-T-P-C : 0-0-4-2

Intended for : BS Chemical Sciences

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Module I: Data Analysis and Programming**

1. Determination of distribution, mean, variance, standard deviation, confidence interval from sample data obtained for the space variables of diffusive particles at a finite-time.
2. Determination of correlations, auto-correlations and spectral density of sample data obtained for the space and momentum variables of diffusive particle at a given time.
3. Linear and non-linear curve fitting (regression analysis) of given spectroscopic data (Abs. coefficient. vs. T/FI. Decay) and determination of physical properties from fitting.
4. Writing a Fortran/C++ program for matrix multiplication, diagonalization, and calculation of roots of a Secular determinant.

- **Module II: Physical Properties**

1. Determining a given solution's viscosity coefficient with Ostwald's viscometer considering aqueous solutions of glycerol, ethanol, etc.
2. Determination of surface tension of a given solution by drop weight method using a stalagmometer, considering aqueous solutions of NaCl, acetic acid, ethanol, etc., as systems.

- **Module III: Chemical Kinetics**

1. Study of kinetics of saponification of ester by using the conductometric method.
2. Study of the kinetics of the reaction $I^- + S_2O_8^{2-}$ by colorimetric method.
3. Acid hydrolysis of methyl acetate at different temperatures at a given concentration of $[H^+]$ ions.

- **Module IV: Spectroscopy**

1. Verification of Beer-Lambert law using colorimetry.
2. Absorption spectrum of a conjugated dye, polymethine, interpretation of the spectra using the "free-electron" model.
3. Determination of quantum yield.

- **Module V: Electrochemistry**

1. Determination of emf of an electrochemical cell and measurement of thermodynamic parameters from the temperature dependence of emf.
2. Determination of E^0 of Fe^{3+}/Fe^{2+} couple in the hydrogen scale by potentiometric titration of ferrous ammonium sulfate solution using $KMnO_4$, or $K_2Cr_2O_7$ as standard.



Textbooks:

1. D. P. Shoemaker, C. W. Garland, and J. W. Nibler, **Experimental Physical Chemistry**, 8th Edition, McGraw Hill, 2009.
2. G. Peter Matthews, **Experimental Physical Chemistry**, Oxford University Press, 1986.

References:

1. Frederick A. Bettelheim, **Experimental physical chemistry**, Saunders, 1971.
2. A. Ghosal, B. Mahapartra, A. K. Nad, **An Advanced Course in Practical Chemistry**, New Central Book Agency Pvt Ltd, 2000.